WASP-43b

R-1031

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_160422` · created 2026-07-30T06:18:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457500.73 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.026 +/- 0.057
Transit depth (Rp/Rs)^2	0.069 +/- 0.3 [%]
Orbital Inclination (inc)	77.68 +/- 2.92
Airmass coefficient 1 (a1)	0.91 +/- 0.43
Airmass coefficient 2 (a2)	0.033 +/- 0.093
Scatter in the residuals of the lightcurve fit is	45.48 %
Best Comparison Star	#1 - [283, 329]
Optimal Aperture	4.29
Optimal Annulus	4.46
Transit Duration (day)	0.016 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.026 ± 0.057 vs 0.1594 (deviation 2.07σ)
Duration fit vs published	0.016 d vs 0.0512 d (deviation 1.84σ)
Mid-time O–C	0.2 min
Depth SNR	0.4
Residual scatter	45.48 %

Light curve

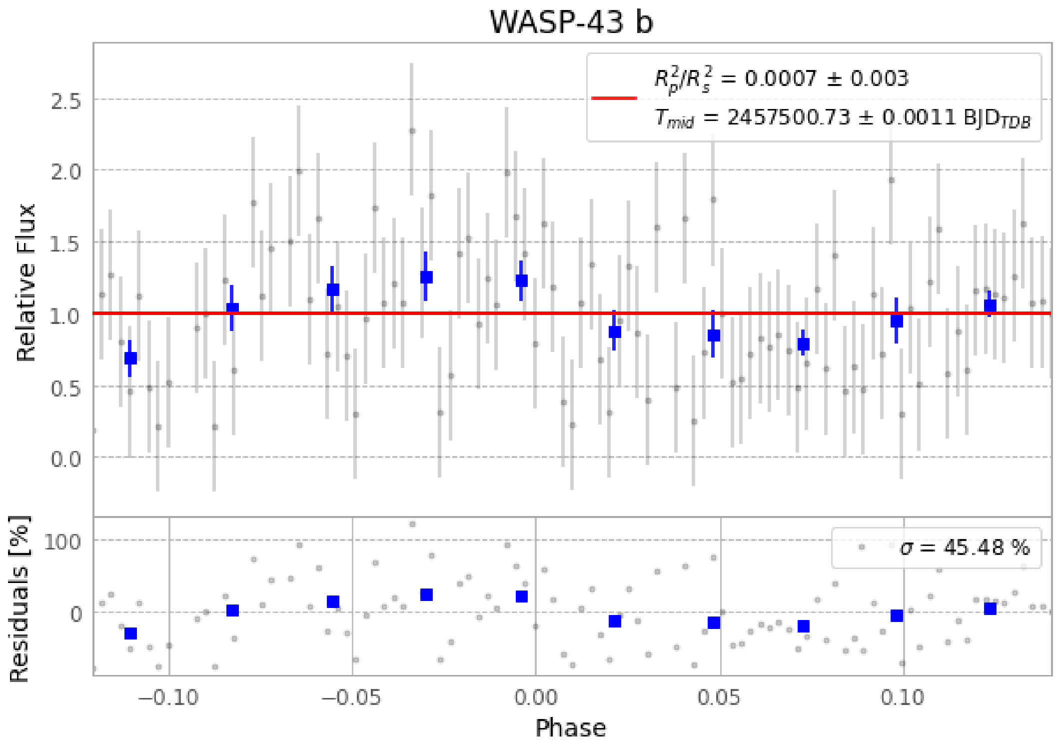

Figure 1 — FinalLightCurve_WASP-43 b_2016-04-21.png (SHA-256 28867ecaf56d73a2…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 158], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 da45673c673e82f3…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b65555c`

Data provenance

103 FITS frames (68 MB), `WASP-43160422030312.FITS` → `WASP-43160422080912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160422.log (5a36233e25f6…), FinalLightCurve_WASP-43 b_2016-04-21.png (28867ecaf56d…), FinalLightCurve_WASP-43 b_2016-04-21.csv (d63b6095c729…)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)